Piyush Garg

Delhi NCR | erpiyushgarg17@gmail.com | GitHub | LinkedIn | Portfolio

Education

Bachelor of Technology in AI & Data Science Expected Graduation: May 2028
Akhilesh Das Gupta Institute of Professional Studies *Current CGPA: 7.8/10.0*

Relevant Coursework: Deep Learning, Data Structures & Algorithms, Calculus, Probability

Central Board of Secondary Education (CBSE 12th) Graduated: June 2024
Bal Bhavan International School *Score: 75%*

Skills

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Generative Models (GANs, Diffusion, Autoencoders)

Programming: Python, C++, C, JavaScript

Backend & AI Systems: FastAPI

Scientific Computing: NumPy, Pandas, Matplotlib

Languages: English, Hindi

Projects

TriageIQ — AI-Powered Insurance Claims Triage System (In Progress) GitHub

- Building a multimodal AI pipeline that processes insurance claims from PDFs, images, audio, and text using FastAPI and HuggingFace Transformer models.
- Developing asynchronous ML workflows with named entity recognition (NER), zero-shot classification, semantic similarity search, and fraud-signal analysis.
- Implementing pgvector-based retrieval, vector embeddings, and Retrieval-Augmented Generation (RAG) pipelines for policy-document querying and contextual reasoning.
- Designing scalable backend architecture and modular AI services with planned LangGraph multi-agent orchestration for automated claim analysis and intelligent reporting.

SynapseAI — Knowledge Graph & Reasoning System (In Progress) GitHub

- Developing a structured knowledge graph system that converts unstructured text into concepts and relationships using LLM-based extraction pipelines.
- Implementing graph traversal, multi-hop reasoning, relationship querying, and persistent graph storage for efficient semantic exploration and contextual retrieval.
- Designing modular NLP and reasoning workflows focused on scalable AI memory systems, graph-based reasoning, and future RAG-based integration.
- Building local-first knowledge architectures for structured information retrieval, semantic reasoning, and intelligent concept linking across connected entities.

Diffusion-Based Generative Modeling for Sparse Detector Data GitHub

- Implemented a denoising diffusion probabilistic model (DDPM) pipeline in PyTorch for sparse calorimeter detector event generation.
- Designed timestep-conditioned denoising workflows, data preprocessing pipelines, and physics-aware evaluation strategies for generative modeling of detector showers.
- Explored sparse detector representations, diffusion sampling techniques, and high-dimensional data generation under constrained computational settings.
- Investigated architectural and computational bottlenecks affecting training stability, reconstruction fidelity, and scalability of diffusion-based learning systems.